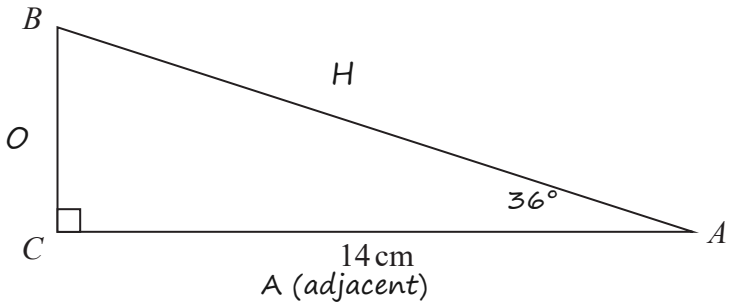
25 ABC is a right-angled triangle.



$AC = 14$ cm.
Angle $C = 90°$

size of angle B : size of angle $A = 3 : 2$

Work out the length of AB.
Give your answer correct to 3 significant figures.

Angles A and B total 180° − 90° = 90°.

	Parts	Degrees	
÷5	5	90	÷5
×2	1	18	×2
	2	36	

1 mark
A has 2 of the 5 ratio parts so angle A = 36°

$$\cos 36° = \frac{14}{AB}$$

1 mark
Adjacent ÷ hypotenuse

$$AB = \frac{14}{\cos 36°} = 17.30495168...$$

1 mark
Rearrange before calculating

AB = 17.3 cm (3 significant figures)

1 mark
Round only the final length

17.3 cm

(Total for Question 25 is 4 marks)